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A Project Report on

"MULTIMODAL DEEPFAKE AND SYNTHETIC MEDIA DETECTOR"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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CERTIFICATE

This is to certify that the project work titled **"MULTIMODAL DEEPFAKE AND SYNTHETIC MEDIA DETECTOR"** is carried out by **AKASH SM (23BTRCL244), AVINASH G K (23BTRCL248), ADITYA K (23BTRCL127), ARYAN AGRAWAL (23BTRCL131), DEVANSH GUPTA M (23BTRCL135), DRUVRAAJ (23BTRCL137)** bonafide students of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfilment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

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DECLARATION

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Signature of Student(s)

TABLE OF CONTENTS

Front Matter

Certificate
Declaration
Acknowledgement
Abstract
List of Figures

CHAPTER 1: INTRODUCTION

1.1 Background of Deepfake Technology
1.2 Impact and Risks
1.3 Limitations of Existing Methods
1.4 Motivation for Multimodal Detection
1.5 Aim of the Project

CHAPTER 2: LITERATURE SURVEY

2.1 Overview of Deepfake Detection
2.2 CNN-Based Approaches
2.3 Xception-Based Models
2.4 Multimodal Detection Techniques
2.5 Comparative Analysis
2.6 Research Gap

CHAPTER 3: PROPOSED SYSTEM

3.1 Problem Statement
3.2 Key Challenges
3.3 Proposed Architecture
3.4 System Workflow

CHAPTER 4: DATASET AND PREPROCESSING

4.1 Dataset Overview
4.2 FakeAVCeleb Dataset
4.3 VoxCeleb2 Dataset
4.4 Data Balancing
4.5 Preprocessing Pipeline

CHAPTER 5: SYSTEM ARCHITECTURE

5.1 Overall Architecture
5.2 Visual Branch (Xception)
5.3 Audio Branch (CNN)
5.4 Cross-Modal Attention Fusion

CHAPTER 6: METHODOLOGY AND TRAINING

6.1 Training Strategy
6.2 Class Imbalance Handling

6.3 Data Augmentation
6.4 Optimization Techniques

CHAPTER 7: IMPLEMENTATION
7.1 Tools and Technologies
7.2 System Pipeline
7.3 Hardware Requirements

CHAPTER 8: RESULTS AND DISCUSSION
8.1 Evaluation Metrics
8.2 Model Performance
8.3 Confusion Matrix
8.4 ROC Curve
8.5 Observations

CHAPTER 9: FUTURE WORK
9.1 Limitations
9.2 Future Improvements

CHAPTER 10: CONCLUSION
10.1 Summary
10.2 Key Contributions

REFERENCES

ABSTRACT

The rapid advancement of deep learning has enabled the creation of highly realistic synthetic media, commonly referred to as deepfakes, where both facial appearance and voice can be convincingly manipulated within video content. The increasing accessibility of such technologies poses serious threats to digital trust, misinformation control, and personal security, necessitating the development of robust and reliable detection systems. Traditional deepfake detection approaches predominantly rely on visual analysis alone, making them insufficient for detecting modern multimodal deepfakes that simultaneously manipulate both visual and audio streams.

This project presents a multimodal deepfake detection system that jointly analyzes visual and audio information to improve detection robustness and generalization. The proposed architecture consists of a dual-branch framework: a visual branch based on the pretrained Xception network for extracting spatial facial features, and an audio branch based on a convolutional neural network processing Mel-spectrogram and MFCC representations. A key contribution of this work is the introduction of a cross-modal attention fusion mechanism, which dynamically calibrates the contribution of each modality on a per-sample basis, effectively suppressing unreliable signals and enhancing overall prediction quality.

The system is trained and evaluated on the FakeAVCeleb dataset, supplemented with real samples from VoxCeleb2 to address severe class imbalance. A rigorous evaluation protocol based on identity-level data splitting is employed to eliminate identity leakage and ensure genuine generalization. The final model achieves 94% accuracy and a fused ROC-AUC of 0.981 on unseen validation data.

The results demonstrate that multimodal learning combined with attention-based fusion significantly improves deepfake detection performance compared to unimodal approaches, contributing toward reliable real-world media authentication systems.

LIST OF FIGURES

Sr. No.	Figure Title	Page No.
1	System Architecture of Multimodal Deepfake Detection Model	
2	Cross-Modal Attention Fusion Mechanism	
3	Project Structure	
4	ROC-AUC Comparison Chart	
5	Precision-Recall Score Chart	

Introduction

The rapid advancement of deep learning has enabled the creation of highly realistic synthetic media — commonly referred to as deepfakes — in which a person's face, voice, or both can be convincingly manipulated or entirely replaced within video content. What was once a computationally expensive and technically demanding process has become increasingly accessible, with modern generative models capable of producing deepfake videos that are difficult or impossible to distinguish from genuine recordings with the naked eye. This democratisation of deepfake generation has introduced serious and growing threats to digital trust, personal security, and the integrity of information across journalism, law enforcement, political discourse, and social media platforms.

The consequences of undetected deepfakes are significant. Fabricated video evidence can be used to defame individuals, manipulate public opinion, undermine legal proceedings, and spread disinformation at scale. As the quality and availability of deepfake generation tools continue to improve, the need for reliable, automated detection systems has become an urgent research priority.

Existing detection approaches have predominantly focused on the visual stream alone, analysing facial inconsistencies introduced by face swapping or neural rendering techniques. However, modern deepfakes increasingly manipulate both the visual and audio components of a video simultaneously — synthesising faces and cloning voices together — meaning that visual-only detectors are inherently limited in their coverage of the full manipulation landscape.

This project addresses that gap by developing a multimodal deepfake detection system that jointly analyses both the visual and audio streams of a video. By treating each modality as an independent source of evidence and combining them through a principled fusion mechanism, the system is designed to detect a broader range of manipulation types than any single-modality approach could achieve.

The system was trained and evaluated on FakeAVCeleb, a purpose-built multimodal deepfake benchmark covering four distinct manipulation categories, supplemented with real video data from VoxCeleb2 to address the dataset's severe class imbalance. Development was guided throughout by rigorous evaluation methodology, including strict identity-level data splitting to ensure that reported performance reflects genuine generalisation to unseen subjects rather than memorisation of training identities.

Aim of the Project

The overarching aim of this project is to design, implement, and evaluate a robust multimodal deepfake detection system capable of identifying manipulated video content by jointly analysing visual and audio evidence. The project pursues this aim through the following specific objectives:

- **Develop a dual-branch detection architecture** that processes visual and audio streams independently through specialised neural network branches — an Xception-based visual branch and a CNN-based audio branch — before combining their representations through a learned fusion mechanism.
- **Design a cross-modal attention gate** that dynamically calibrates the contribution of each modality on a per-sample basis, preventing a weak or noisy modality from degrading the quality of the fused prediction.
- **Address severe class imbalance** present in the FakeAVCeleb dataset through a multi-layered strategy combining balanced batch sampling, per-sample class weighting, and focal loss, supplemented by real video data sourced from VoxCeleb2.

- **Establish a rigorous evaluation protocol** using strict identity-level data splitting to ensure that no subject appears in both training and validation sets, producing honest performance metrics that reflect genuine generalisation rather than identity leakage.
- **Develop a stable training strategy** through a two-phase training protocol that preserves pretrained visual representations during early fusion head initialisation before transitioning to full end-to-end fine-tuning.
- **Improve real-world robustness** through targeted data augmentation including JPEG compression simulation for the visual branch and SpecAugment-style masking for the audio branch, ensuring the model generalises to video content that has been re-encoded through social media platforms.
- **Evaluate and quantify the contribution of each component** through an ablation study comparing different training configurations, data strategies, and architectural choices on the same held-out identity-level evaluation set.

<table>
<tr><th>SL.NO</th><th>TITLE</th><th>LITERATURE SURVEY</th></tr>
<tr>
<td>1.</td>
<td>Advancing Deepfake Detection Using Xception Architecture: A Robust Approach for Safeguarding against Fabricated News on Social Media

[Dunya Ahmed Alkurdi , December 2024]</td>
<td>

1. Introduction

Deepfake technology has become a major threat by creating highly realistic fake images and videos, especially on social media. This paper focuses on detecting such manipulated content using the **Xception architecture**, which is effective in extracting deep visual features. The goal is to improve detection accuracy and ensure authenticity of digital media.

2. Existing Work

- Earlier methods used **CNNs, RNNs, and capsule networks** for deepfake detection
- Some approaches focused on:
  - Facial artifacts
  - Temporal inconsistencies
  - Frequency-domain features
- Models like **MesoNet, ResNet, and LSTM-based systems** were used
- However, these methods faced issues with:
  - Generalization
  - Detection of advanced deepfakes

3. Methodologies Used in Existing Studies

- **CNN-based approaches:** Extract spatial features from images
- **RNN/LSTM models:** Capture temporal changes in videos
- **Capsule Networks:** Detect hierarchical relationships in facial features
- **Frequency-based methods:** Analyze inconsistencies in image frequency domain
- **Hybrid models:** Combine multiple techniques for better accuracy

These methods aim to detect deepfake artifacts but often struggle with unseen manipulations and real-world variations

Accuracy	Precision	Recall	F1 score	AUC-ROC
0.9969	0.9958	0.9980	0.9969	0.9999

4. Conclusion:

The paper concludes that the **Xception-based model** provides highly accurate deepfake detection (up to ~99.65%). It highlights that strong feature extraction and proper preprocessing significantly improve performance. The study emphasizes the need for real-time detection systems

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		and further improvements to handle evolving deepfake techniques and ensure secure digital environments.
2.	**Deepfake Detection Using XceptionNet** [Muskan Kumari, Sanya Gupta, Neha Singh May 2025]	1. Introduction The rapid advancement of artificial intelligence has led to the emergence of deepfake technology, which enables the creation of highly realistic manipulated videos. These deepfakes pose serious threats such as misinformation, identity fraud, and loss of trust in digital media. The paper focuses on detecting deepfake videos using deep learning techniques, specifically the XceptionNet architecture. The study utilizes the FaceForensics++ dataset, which contains over 1.8 million manipulated images generated using multiple techniques such as FaceSwap, Face2Face, NeuralTextures, and DeepFakes. The objective is to classify images as real or fake using facial feature analysis and deep convolutional neural networks. 2. Existing Work Previous research in deepfake detection includes both traditional and deep learning-based approaches: Traditional Forensics Methods: Use handcrafted features and statistical analysis (e.g., steganalysis + SVM), but perform poorly under compression. CNN-Based Models: Models like MesoNet and XceptionNet automatically learn features and perform better than traditional methods. Face Manipulation Techniques: Methods such as Face2Face, Deep Video Portraits, and NeuralTextures enable realistic facial modifications. GAN-Based Approaches: Generative Adversarial Networks (GANs) are widely used for creating high-quality fake images and videos. Detection Techniques: Focus on inconsistencies like eye blinking, texture artifacts, and facial distortions. Limitations: Many systems fail under real-world conditions like compression, resizing, and social media processing. 3. Methodologies Used The paper proposes a deep learning-based methodology using XceptionNet: Dataset Used: FaceForensics++ dataset with large-scale manipulated video frames.

		Preprocessing: Face detection and cropping Enlargement of facial region (1.3× scaling) Focus on relevant facial features Model Used: XceptionNet (pretrained on ImageNet) Modified final layer for binary classification (real vs fake) Training Strategy: Fine-tuning on dataset Evaluation under different video qualities: Raw (uncompressed) HQ (high quality compressed) LQ (low quality compressed) Evaluation Metrics: Accuracy Precision Robustness under compression **Performance** Accuracy: >95% Precision: Not specified Recall: Not specified F1 Score: Not specified 4. Conclusion The study demonstrates that XceptionNet is highly effective for deepfake detection, achieving over 95% accuracy on high-quality videos and maintaining above 80% accuracy under heavy compression. The model outperforms both traditional techniques and human detection capabilities.
3.	**DEEPFAKE DETECTION BASED ON THE XCEPTION MODEL** [January 2022]	**1. Introduction** Deepfake technology uses AI to create realistic fake videos by replacing faces. With the rapid growth of social media, deepfakes have become a serious threat, causing misinformation and security issues. This paper focuses on detecting deepfakes using **Xception-based deep learning models**, which are effective in extracting facial features and identifying manipulations. **2. Existing Work** Traditional deepfake detection methods relied on media forensics techniques such as analyzing camera fingerprints and detecting image artifacts introduced during manipulation. These approaches focused on identifying inconsistencies in optical lens patterns, color interpolation, and compression traces. However, with the advancement of deepfake generation techniques, these traditional methods became less effective.

		To overcome these limitations, researchers introduced deep learning-based approaches, particularly Convolutional Neural Networks (CNNs), which significantly improved detection accuracy by automatically learning complex visual features. Earlier works mainly focused on detecting face-swapping techniques and GAN-based manipulations. Despite achieving promising results, many of these methods still suffer from limitations such as poor robustness and lack of generalization when applied to unseen or real-world deepfake data. **3. Methodologies Used** Existing works in deepfake detection primarily rely on deep learning and image forensics techniques. Convolutional Neural Networks (CNNs) are widely used to extract spatial features from images, enabling the model to identify subtle visual patterns and inconsistencies. Additionally, high-pass filtering techniques are employed to capture pixel-level anomalies introduced during image manipulation. Many approaches also focus on detecting facial manipulations by analyzing swapped or altered facial regions. Data-driven methods further enhance detection performance by utilizing large-scale datasets such as FaceForensics++, which provide diverse examples of real and fake content. Most of these methods adopt supervised learning, where models are trained using labeled datasets to classify inputs as real or fake. Although these methodologies have improved detection accuracy, they still face challenges in handling advanced deepfake techniques and generalizing well to unseen data. • **CNN-based models:** Extract spatial features from images • **High-pass filtering techniques:** Detect pixel-level inconsistencies • **Face manipulation detection methods:** Identify swapped or altered faces • **Data-driven approaches:** Use large datasets like FaceForensics++ • **Supervised learning:** Train models using labeled real/fake data These methods aim to detect deepfakes but may struggle with advanced manipulations. **Performance** Accuracy: 98.77%

		Precision: Not provided Recall: Not provided F1 Score: Not provided **4. Conclusion** The paper concludes that the **Xception model** is highly effective for deepfake detection, achieving very high accuracy (~98.77%). It highlights that combining **face detection (MTCNN) with Xception** improves performance. The study also emphasizes the need for more robust models to handle evolving deepfake techniques in real-world scenarios.
4.	**Xception: Deep Learning with Depthwise Separable Convolutions** [Francois Chollet]	**1. Introduction** Convolutional Neural Networks (CNNs) have become the dominant approach in computer vision tasks such as image classification. Over time, architectures like AlexNet, VGG, and Inception have improved performance by increasing depth and optimizing feature extraction. However, traditional convolution operations simultaneously capture spatial and cross-channel correlations, which can be computationally expensive. This paper introduces **Xception (Extreme Inception)**, a novel CNN architecture proposed by François Chollet. The key idea is to replace standard Inception modules with **depthwise separable convolutions**, which separate spatial and channel-wise feature learning. This results in a more efficient use of parameters while maintaining or improving performance. The proposed Xception model demonstrates better performance than Inception V3 on large-scale datasets like ImageNet and JFT, showing that depthwise separable convolutions can be a powerful alternative in deep learning architecture design. **2. Existing Work** Earlier research in computer vision primarily focused on Convolutional Neural Network (CNN) architectures such as LeNet and AlexNet, which used stacked convolution and pooling layers for feature extraction. These were later improved by deeper networks like VGG, which increased model depth to learn more complex features. A major advancement came with the Inception architecture, which introduced parallel convolutional filters of different sizes to capture diverse spatial features efficiently. Inception models improved performance while reducing computational cost compared to traditional CNNs. Additionally, research on **depthwise separable convolutions** explored separating spatial and channel-wise feature extraction to make models more efficient.

		The Xception architecture builds upon these previous works by combining the strengths of Inception modules and depthwise separable convolutions, aiming to further improve performance and efficiency in deep learning models. **3. Methodologies Used** The Xception architecture is based on the concept of **depthwise separable convolutions**, which divide the standard convolution operation into two separate steps: depthwise convolution (spatial feature extraction) and pointwise convolution (channel-wise feature mapping). This separation improves computational efficiency and allows better feature learning. The network consists of **36 convolutional layers organized into 14 modules**, with most modules connected using **residual connections** to improve training stability and convergence. The architecture is structured into three main flows: entry flow, middle flow, and exit flow, which progressively extract higher-level features from the input images. Batch normalization is applied after each convolution layer to stabilize learning, and a final **logistic regression layer** is used for classification. The model is trained using large-scale datasets such as ImageNet and JFT, with optimization techniques like stochastic gradient descent and RMSProp. **Performance** Top-1 Accuracy: 79.0% Top-5 Accuracy: 94.5% Precision: Not specified Recall: Not specified F1 Score: Not specified **4. Conclusion** The paper concludes that the proposed **Xception architecture** is an effective and efficient deep learning model for image classification tasks. By replacing traditional convolution operations with **depthwise separable convolutions**, the model achieves better utilization of parameters and improved performance. The use of **residual connections** further enhances training stability and convergence. Experimental results show that Xception slightly outperforms Inception V3 on the ImageNet dataset and achieves significantly better results on the large-scale JFT dataset. This demonstrates that separating spatial and channel-wise feature extraction can lead to more powerful representations. Overall, the study highlights that Xception is a simple yet highly effective architecture and suggests that depthwise separable convolutions will play an important role in future deep learning models.

Proposed System and Methodology

Introduction

Deepfake technology leverages deep learning algorithms to synthesise or manipulate video content with unprecedented realism. The ability to convincingly replace a person's face, voice, or both in video recordings poses serious threats to information integrity, digital trust, and personal security. As deepfake generation techniques have grown increasingly sophisticated, automated detection has become an urgent and critical research challenge with wide-ranging implications for journalism, law enforcement, social media platforms, and public discourse.

This report presents the design, architecture, training strategy, and evaluation results of a multimodal deepfake detection system that jointly analyses both the visual and audio streams of a given video.

The system advances beyond single-modality approaches by combining visual and audio signals through a cross-modal attention fusion mechanism, enabling the model to adaptively weight each modality based on its reliability for any given input.

The proposed system was trained and evaluated on the FakeAVCeleb dataset — a comprehensive multimodal deepfake benchmark — supplemented with real videos sourced from VoxCeleb2 to address the severe class imbalance inherent in the original dataset.

Through careful architectural design and rigorous evaluation methodology, the final model achieves 94% accuracy and a fused ROC-AUC of 0.981 on an identity-level held-out split comprising 2,846 samples from subjects never encountered during training, demonstrating robust generalisation beyond the training distribution.

Key Contribution: A cross-modal attention gate that dynamically suppresses unreliable modalities on a per-sample basis,
combined with a two-phase training strategy and multi-layered class imbalance handling, achieving 0.981 ROC-AUC
with 0.99 fake recall on unseen identities.

1. Problem Statement

Deepfake detection is not a straightforward binary classification task. The detection of deepfake multimedia content presents several distinct and interconnected technical challenges that must be addressed simultaneously. This section enumerates each challenge and explains why naive approaches fail in practice.

1.1 Class Imbalance

FakeAVCeleb contains approximately 500 genuine videos against over 21,000 synthetic videos — a 42:1 imbalance ratio. A naive classifier that predicts everything as fake trivially achieves 97% accuracy on this dataset, yet is completely useless for detecting real videos.

Standard binary cross-entropy loss in this setting collapses training toward the majority (fake) class with no incentive to learn real-video features.

1.2 Identity Leakage

If the train-validation split is performed at the video level rather than the identity level, the same person may appear in both the training and validation sets. When this occurs, the model learns to perform face

recognition rather than deepfake detection, identifying whether a person is 'known' from training rather than detecting manipulation artifacts.

This produces severely inflated validation metrics — naive video-level splits in this project yielded 99% accuracy, which was confirmed as identity leakage, not genuine deepfake detection.

1.3 Weak Audio Modality

Audio-based deepfake artefacts are considerably subtler and more difficult to learn than visual artefacts. Voice cloning produces spectrogram representations that closely resemble genuine speech, making it difficult for a shallow model to distinguish synthesised audio.

A naively trained audio branch without specialised architecture and augmentation produces near-random predictions, and introducing such a branch into a fusion layer actively injects noise that degrades overall performance.

1.4 Fusion Instability

Simple feature concatenation assigns equal weight to both modalities with no mechanism to suppress an unreliable branch. When the audio branch is weak or noisy, equal-weight fusion means the strong visual branch becomes corrupted by audio noise rather than benefiting from any complementary signal. A principled fusion mechanism is required that can learn to calibrate each modality's contribution per sample.

1.5 Training Instability

Fine-tuning a large pretrained convolutional network such as Xception simultaneously with randomly initialised fusion layers introduces severe gradient instability.

The large gradients from the untrained layers overwrite pretrained ImageNet representations before they can be adapted to deepfake detection. In practice, this manifested as binary cross-entropy loss values consistently above 4.0 — indicating maximum model uncertainty — until a phased training strategy was adopted.

Technical challenges encountered during development and the corresponding targeted solutions implemented

Challenge	Root Cause	Solution Adopted
Class Imbalance	42:1 fake-to-real ratio	Balanced batches + focal loss + sample weights
Identity Leakage	Video-level splitting	GroupShuffleSplit by identity regex (id#####)
Training Instability	Random fusion layers corrupting Xception	Two-phase training (freeze Xception in Phase 1)
Weak Audio Signal	Shallow CNN, no augmentation	4-block CNN with SeparableConv2D + SpecAugment
Fusion Instability	Equal-weight concatenation	Cross-modal attention gate (per-sample weighting)
Real-World Brittleness	Training on pristine video only	JPEG compression + SpecAugment augmentation

Real Recall Collapse	Focal loss alone insufficient for 42:1	VoxCeleb2 real supplement + 4:1 undersampling cap

2. Dataset

The proposed system was trained and evaluated using two complementary datasets: FakeAVCeleb, a purpose-built multimodal deepfake benchmark, and VoxCeleb2, a large-scale speaker recognition corpus used to supplement the real video class. This section describes each dataset, the rationale for their combination, and the identity-level data splitting methodology employed to ensure evaluation validity.

2.1 FakeAVCeleb

FakeAVCeleb is a multimodal deepfake dataset specifically designed for audio-visual deepfake detection research. It contains videos drawn from VoxCeleb2 source identities and systematically manipulated using state-of-the-art face swapping and voice cloning techniques. The dataset covers four distinct manipulation categories, each representing a different combination of real and synthetic visual and audio content:

- Real Video – Real Audio (RVRA): Genuine unmodified video and audio, constituting the real class.
- Fake Video – Fake Audio (FVFA): Fully synthesised content with both face swapping and voice cloning applied.
- Fake Video – Real Audio (FVRA): Face-swapped video paired with the original unmodified audio track.
- Real Video – Fake Audio (RVFA): Genuine face video with a voice-cloned audio replacement.

Dataset statistics at each stage of data pipeline construction.

Split	Real Videos	Fake Videos	Total	Ratio (Fake:Real)
FakeAVCeleb Raw	500	21,066	21,566	42:1
After Pre-processing	499	21,064	21,563	42:1
Training Pool (with VoxCeleb2)	~4,567	~18,268	~22,835	4:1

The most significant characteristic of FakeAVCeleb is its extreme class imbalance: only approximately 500 genuine videos exist against over 21,000 synthetic examples.

This 42:1 ratio makes naive classifier training unreliable, as a model predicting everything as fake achieves 97.7% accuracy whilst providing zero utility for detecting genuine content.

2.2 VoxCeleb2 Supplement

To address the critical scarcity of real video samples in FakeAVCeleb, approximately 4,067 genuine videos were sourced from VoxCeleb2. VoxCeleb2 is a large-scale speaker recognition dataset comprising over one million video clips of celebrities in real-world, in-the-wild conditions collected from interview footage, talk shows, and public appearances.

VoxCeleb2 is a particularly appropriate supplement because the FakeAVCeleb synthetic videos were originally generated using VoxCeleb2 source identities. This makes VoxCeleb2 genuine videos semantically consistent with the FakeAVCeleb domain in terms of visual quality, recording conditions,

and subject characteristics. The combined dataset brings the total real sample count to approximately 4,567, enabling training at a more manageable 4:1 fake-to-real ratio whilst still retaining approximately 87% of the full FakeAVCeleb fake set.

2.3 Identity-Level Data Splitting

One of the most critical methodological decisions in this project concerns the strategy used to partition data into training and validation sets. An identity-level split is mandatory for honest evaluation of deepfake detection performance.

If the split is performed at the video level, the same person may appear in both training and validation sets. In this scenario, the model learns face identity rather than deepfake artefacts, leading to severely inflated evaluation metrics. This was empirically confirmed in this project: a naive video-level split produced 99% validation accuracy, whereas the correct identity-level split produced 94-95% — a difference of five to six percentage points attributable entirely to data leakage.

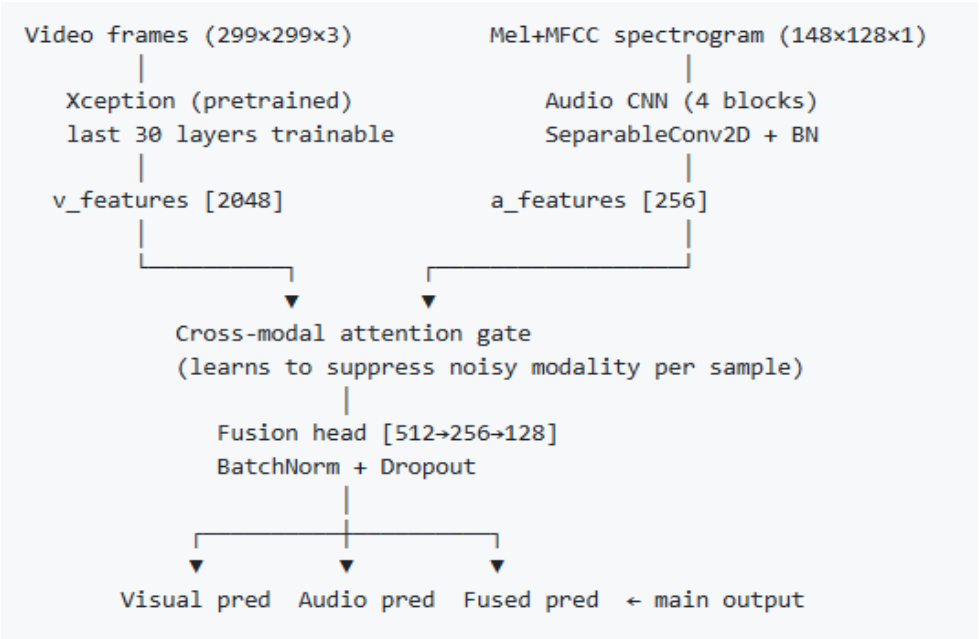
The correct approach uses GroupShuffleSplit with groups defined at the identity level. Identity labels are extracted from filenames using the regular expression pattern id##### (e.g., id07161 from id07161_xyz.mp4). This guarantees that all videos belonging to a given person fall entirely within either the training set or the validation set, but never both. An 80:20 train-to-validation split ratio was applied.

Impact of data splitting strategy on validation accuracy and result validity.

Approach	Val Accuracy	Identity Overlap	Validity
Video-level split (naive)	99%	100% overlap	INVALID — identity leakage
Identity-level split (correct)	94–95%	0% overlap	VALID — genuine result

3. System Architecture

The proposed system follows a multimodal dual-branch architecture that processes video and audio streams independently before combining them through a cross-modal attention fusion mechanism. The pipeline comprises three principal components: a visual branch based on Xception, an audio branch based on a four-block CNN, and a fusion module with three parallel output heads. The architecture is illustrated below.



3.1 Visual Branch

The visual branch is responsible for extracting high-level facial feature representations that encode deepfake artefacts present in manipulated face regions. The input to the visual branch is a 299x299x3 pixel face crop extracted from each video frame using the MTCNN (Multi-task Cascaded Convolutional Networks) face detector. MTCNN is employed for its robustness to varying pose, illumination, and scale conditions.

The face crop is passed through Xception, a deep convolutional network architecture pretrained on the ImageNet large-scale image classification benchmark. Xception was selected as the visual backbone due to its use of depthwise separable convolutions that efficiently capture fine-grained spatial features and subtle textural artefacts introduced by face synthesis techniques such as neural rendering and GAN-based face swapping.

To preserve the rich pretrained representations while adapting to the deepfake detection domain, the first layers of Xception are frozen during initial training. Only the last 30 layers remain trainable in Phase 1 of the training pipeline. The output of the Xception backbone passes through a Global Average Pooling layer and a Dropout layer to produce a compact 2,048-dimensional visual feature vector.

Visual branch component specifications.

Component	Detail
Input Shape	299 x 299 x 3 (face crop, RGB)
Backbone	Xception (pretrained on ImageNet)
Trainable Layers — Phase 1	Last 30 layers only; earlier layers frozen
Trainable Layers — Phase 2	All layers unfrozen for end-to-end fine-tuning
Pooling	Global Average Pooling
Regularisation	Dropout applied after pooling
Output Dimension	2,048-dimensional feature vector

3.2 Audio Branch

The audio branch processes the audio stream of each video to extract temporal and spectral representations that encode synthesised voice artefacts. The input to the audio branch is a stacked representation combining a 128-band Mel-spectrogram with 20 Mel-frequency Cepstral Coefficients (MFCCs), concatenated along the feature axis to produce a 148x128x1 tensor.

This dual representation captures both spectral energy distribution across frequency bands and compact vocal tract characteristics encoded in the cepstral domain.

The audio feature tensor is processed by a four-block convolutional neural network designed to progressively extract hierarchical audio representations. The first two blocks employ standard Conv2D layers with Batch Normalisation and Max Pooling to learn low-level and mid-level spectral patterns. Blocks three and four employ SeparableConv2D layers, which provide computational efficiency and act as a regulariser to reduce overfitting on the smaller audio training set. The output is processed through Global Average Pooling and a two-layer projection head to produce a compact 256-dimensional audio feature vector.

Audio branch block-by-block architecture specifications.

Block	Layer Type	Purpose
Block 1	Conv2D + BatchNorm + MaxPool	Low-level spectral feature extraction
Block 2	Conv2D + BatchNorm + MaxPool	Mid-level pattern encoding
Block 3	SeparableConv2D + BatchNorm + MaxPool	Efficient high-level feature extraction
Block 4	SeparableConv2D + BatchNorm + MaxPool	Reduced overfitting via depthwise separation
Projection	GlobalAvgPool + Dense x2	Compact 256-dim audio feature vector

3.3 Cross-Modal Attention Fusion

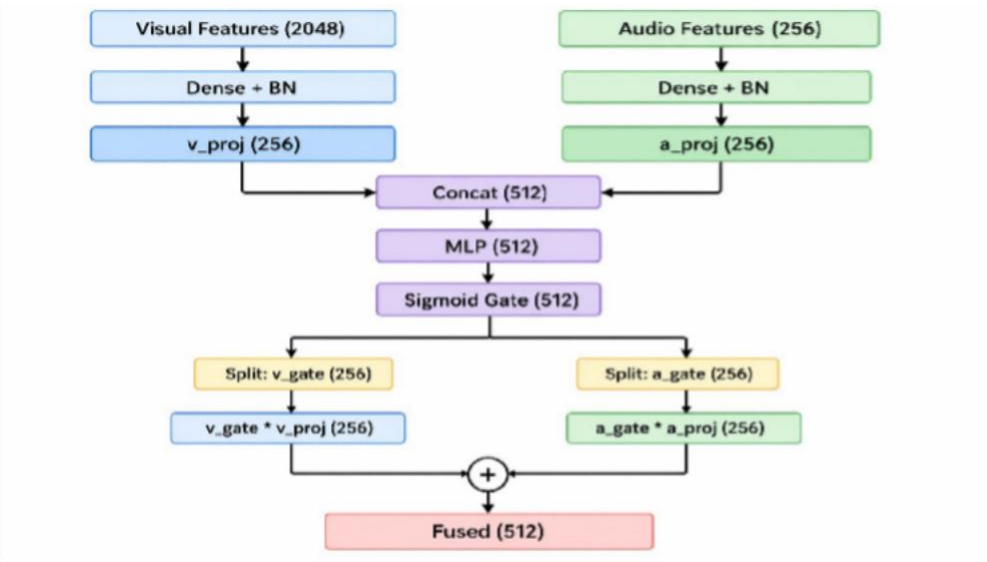
The fusion module is the central architectural innovation of the proposed system. Rather than naively concatenating the visual and audio feature vectors — which assigns equal weight to both modalities regardless of their quality for a given input — the system employs a learned cross-modal attention gate that dynamically calibrates the contribution of each modality on a per-sample basis.

Both the 2,048-dimensional visual feature vector and the 256-dimensional audio feature vector are first projected to a common 256-dimensional representation via dense layers with Batch Normalisation.

The projected representations are concatenated to form a 512-dimensional joint representation, which is processed through a two-layer multi-layer perceptron (MLP) that outputs a 512-dimensional sigmoid gate vector.

This gate vector is split into two 256-dimensional halves — one for the visual projection and one for the audio projection — and applied element-wise to the respective projected features. The resulting gated representations are then concatenated to form the fused 512-dimensional feature vector passed to the fusion head.

The key advantage of this mechanism is that when the audio branch is unreliable or noisy for a particular input, the gate network learns to drive the audio gate weights toward zero, effectively suppressing the audio contribution and preventing noise from corrupting the fused prediction. Conversely, when both modalities carry consistent and complementary signals, the gate allows full contributions from both. This behaviour was empirically validated: the audio branch achieved ROC-AUC of 0.827 — well above random — confirming that the gate extracted meaningful signal whilst appropriately down-weighting it relative to the stronger visual branch at 0.968 AUC.



3.4 Fusion Head and Multi-Output Design

The gated 512-dimensional fused feature vector is processed by the fusion head, comprising three fully connected dense layers of sizes 512, 256, and 128, each followed by Batch Normalisation and Dropout for regularisation. The model produces three simultaneous sigmoid output predictions:

- Visual Only Prediction: Produced directly from the visual branch features (auxiliary output, loss weight = 1.0).
- Audio Only Prediction: Produced directly from the audio branch features (auxiliary output, loss weight = 0.3).
- Fused Prediction: Produced from the gated fusion head (primary output, loss weight = 3.0).

This multi-output auxiliary design serves a critical training function: it forces each branch to independently learn discriminative features for deepfake detection before fusion begins.

Without auxiliary outputs, the stronger visual branch would dominate all gradient flow, and the audio branch would not receive sufficient training signal to develop meaningful representations.

The asymmetric loss weights (3.0 for fused, 1.0 for visual, 0.3 for audio) reflect the relative importance of each output. At inference time, only the fused prediction is used as the final model output.

Multi-output head configuration, loss weights, and inference usage.

Output Head	Input Features	Loss Weight	Used at Inference
Visual Only Prediction	Visual branch 2048-dim vector	1.0	No
Audio Only Prediction	Audio branch 256-dim vector	0.3	No
Fused Prediction	Gated fusion head 512-dim vector	3.0	Yes (primary output)

4. Training Strategy

The training strategy for the proposed system was developed iteratively to address three core challenges: training instability caused by randomly initialised layers, extreme class imbalance, and the need for real-world robustness under compression and noise. The final strategy combines two-phase training, multi-layered imbalance handling, adaptive callbacks, and on-the-fly data augmentation.

4.1 Two-Phase Training

Training proceeds in two distinct phases to prevent randomly initialised fusion layers from corrupting the pretrained Xception visual representations early in training. Without phased training, binary cross-entropy loss values consistently remained above 4.0, indicating maximum model uncertainty, as large gradients from the untrained fusion head propagated backward through Xception and overwrote its ImageNet representations before they could be adapted to deepfake detection.

Two-Phase Training Configuration

Phase	Max Epochs	Frozen Layers	Learning Rate	Primary Purpose
Phase 1	8	All Xception layers except last 30	1e-4	Stabilise fusion head and audio branch
Phase 2	15	None — all layers trainable	3e-5	End-to-end fine-tuning of full model

The effectiveness of this approach is clearly demonstrated in the training logs.

Phase 1 validation accuracy remained near 0.50 throughout — expected behaviour as the fusion head initialises — while the audio branch and fusion layers stabilised.

Phase 2 then produced a characteristic slow but consistent climb from 0.64 to 0.97 accuracy over 14 epochs, a pattern of genuine learning that contrasts starkly with the immediate high accuracy observed in the identity-leaked model configuration.

4.2 Three-Layer Class Imbalance Handling

The 42:1 fake-to-real imbalance in the raw FakeAVCeleb dataset requires a multi-layered approach. No single mechanism is sufficient: focal loss alone with only 499 real samples failed to prevent real recall collapse to 0.52, as empirically confirmed in the ablation study. The final system employs three complementary mechanisms simultaneously.

4.2.1 Balanced Batch Sampling

Each training batch is constructed to contain exactly 50% real and 50% fake samples, regardless of the overall dataset ratio. The custom Data Generator maintains two separate class sample pools and draws equally from both for each batch of size 16 (8 real and 8 fake samples per batch). This eliminates the gradient imbalance that would otherwise cause the model to prioritise fake sample classification.

4.2.2 Sample Weights

On top of balanced batch sampling, real samples receive a 2x loss weight applied at the individual sample level within each batch. This provides additional emphasis on the minority real class, encouraging the model to penalise real-sample misclassification more heavily, even within the already-balanced batch distribution.

4.2.3 Focal Loss

Standard binary cross-entropy loss is replaced with focal loss as the primary training objective. Focal loss introduces two hyperparameters: gamma (focusing parameter) and alpha (class weighting parameter). With gamma set to 2.0, the loss function down-weights easy examples that the model already classifies correctly — the abundant fake samples that are trivially distinguished early in training — and focuses the training signal on hard examples, particularly the minority real class that is inherently more challenging to learn. The alpha value of 0.75 provides additional weighting to the real (minority) class samples, complementing the gamma-based focusing effect.

Class Imbalance Handling Mechanisms

Mechanism	Parameter	Effect
Balanced Batch Sampling	Batch size = 16	50% real / 50% fake regardless of dataset ratio
Sample Weights	Real weight = 2x	Additional per-sample emphasis on minority class
Focal Loss — Alpha	alpha = 0.75	Extra class-level weighting for real samples
Focal Loss — Gamma	gamma = 2.0	Down-weights easy fake examples; focuses on hard real cases

4.3 Training Callbacks

Four training callbacks were employed to optimise the training process, prevent overfitting, and ensure the best-performing model configuration is retained:

- Early Stopping: Monitors validation fused prediction accuracy. Training terminates if no improvement is observed for 5 consecutive epochs in Phase 1, or 8 consecutive epochs in Phase 2. This prevents over-fitting to the training distribution after the validation metric has plateaued.
- ReduceLROnPlateau: Reduces the active learning rate by 50% when the validation loss shows no improvement for 3 consecutive epochs. This enables the optimiser to escape local optima and continue improving model performance on the validation set.
- Model Checkpoint: Saves the current model weights to best_model.h5 whenever a new maximum validation fused prediction accuracy is achieved, ensuring the best model state is preserved across training.
- CSV Logger: Records all training and validation metrics at the end of each epoch to training_log.csv for comprehensive post-training analysis and ablation study construction.

4.4 Data Augmentation

Data augmentation is applied on-the-fly during training only. Validation and test evaluation always use clean, unaugmented data to ensure reproducible and comparable metrics. The augmentation strategy was designed with a specific goal: force the model to detect manipulation from the underlying semantic acial structure and synthesised audio characteristics rather than pristine, low-level GAN pixel artefacts that disappear when video is shared through social media platforms with lossy re-encoding.

4.4.1 Visual Augmentation

- Horizontal Flip (50% probability): Simulates different camera orientations and reduces sensitivity to left-right directional biases in face synthesis artefacts.
- Brightness Jitter (delta -0.1 to +0.1): Simulates varied lighting conditions across recording environments, preventing the model from relying on absolute brightness characteristics of synthesised faces.
- Random Zoom Crop (50% probability): Simulates different camera-to-subject distances, ensuring the model learns manipulation artefacts at varying face scales.
- JPEG Compression Augmentation (50% probability, quality 30–95): The most important augmentation for real-world robustness. Simulates the lossy re-compression applied by social media platforms when videos are uploaded and shared. This forces the model to detect manipulation from facial structure rather than high-frequency GAN artefacts that are destroyed by JPEG compression.

4.4.2 Audio Augmentation (SpecAugment-Style)

- Frequency Masking: Randomly zeroes out contiguous mel frequency bands in the spectrogram, simulating the spectral degradation introduced by lossy audio codecs such as AAC and MP3.
- Time Masking: Randomly zeroes out contiguous temporal segments of the spectrogram, simulating network packet loss or audio dropout conditions encountered in real-world streaming scenarios.
- Gaussian Noise Injection: Adds Gaussian noise to the spectrogram, simulating ambient background noise from microphones or recording environments.

5. Implementation Details

This section documents the complete implementation specification, including framework selection, hyperparameter configuration, and the pre-processing pipeline architecture.

Complete Implementation Specification

Component	Technology / Value
Deep Learning Framework	TensorFlow 2.x / Keras
Visual Backbone	Xception (tf.keras.applications.Xception, ImageNet pretrained)
Face Detection	MTCNN (Multi-task Cascaded Convolutional Networks)
Face Crop Size	299 x 299 x 3 (normalised to [0, 1])
Audio Features	128-band Mel-spectrogram + 20 MFCC coefficients stacked to 148 x 128 x 1
Audio Processing Library	librosa
Optimiser	Adam
Phase 1 Learning Rate	1e-4

Phase 2 Learning Rate	3e-5 (conservative for stable fine-tuning)
Batch Size	16 (8 real + 8 fake per batch, balanced sampling)
Loss Function	Focal Loss (gamma = 2.0, alpha = 0.75)
Data Split Method	GroupShuffleSplit by identity regex id##### — 80:20 ratio
Precomputed Feature Format	.npy files in processed_faces/ and processed_audio/ directories

5.1 Pre-processing Pipeline

All face crops and audio spectrograms are precomputed and saved as NumPy .npy binary files prior to training. This design decision avoids repeated pre-processing overhead during training epochs and ensures high GPU utilisation by eliminating CPU-bound face detection and audio feature extraction from the training loop. The following scripts constitute the pre-processing pipeline:

- precompute_faces.py: Runs MTCNN on each FakeAVCeleb video, detects and extracts the primary face crop, resizes to 299x299 pixels, normalises pixel values to [0, 1], and saves as a .npy file.
- precompute_audio.py: Extracts 128-band Mel-spectrogram features and 20 MFCC coefficients from each FakeAVCeleb audio track using librosa, stacks to 148x128x1, and saves as .npy.
- precompute_faces_vox.py: Equivalent face extraction script operating on the flat-directory VoxCeleb2 video collection.
- precompute_audio_vox.py: Equivalent audio extraction script for VoxCeleb2 videos, saving output to the same directories.
- Timeout Handling: Each video is processed in a subprocess with a 30-second timeout using multiprocessing.get_context('spawn') to prevent MTCNN from hanging on corrupted or edge-case video frames. Both face and audio scripts are idempotent — already-processed files are automatically skipped on restart.

5.2 Project Structure

The codebase is organised into specialised modules with clear separation of concerns between pre-processing, model definition, training, and evaluation:

```
├── model_fusion.py            # Model architecture + focal loss
├── training.py                # Two-phase training pipeline + DataGenerator
├── evaluate.py                # Deterministic evaluation with threshold sweep
├── preprocess_video.py        # Face extraction (MTCNN) + augmentation
├── preprocess_audio.py        # Mel+MFCC extraction + augmentation
├── precompute_faces.py        # Batch face extraction for FakeAVCeleb
├── precompute_audio.py        # Batch audio extraction for FakeAVCeleb
├── precompute_faces_vox.py    # Batch face extraction for VoxCeleb2
├── precompute_audio_vox.py    # Batch audio extraction for VoxCeleb2
├── train_requirements.txt     # Python dependencies
├── processed_faces/           # Precomputed face arrays (not tracked in git)
├── processed_audio/           # Precomputed audio arrays (not tracked in git)
└── best_model.h5              # Saved model weights (not tracked in git)
```

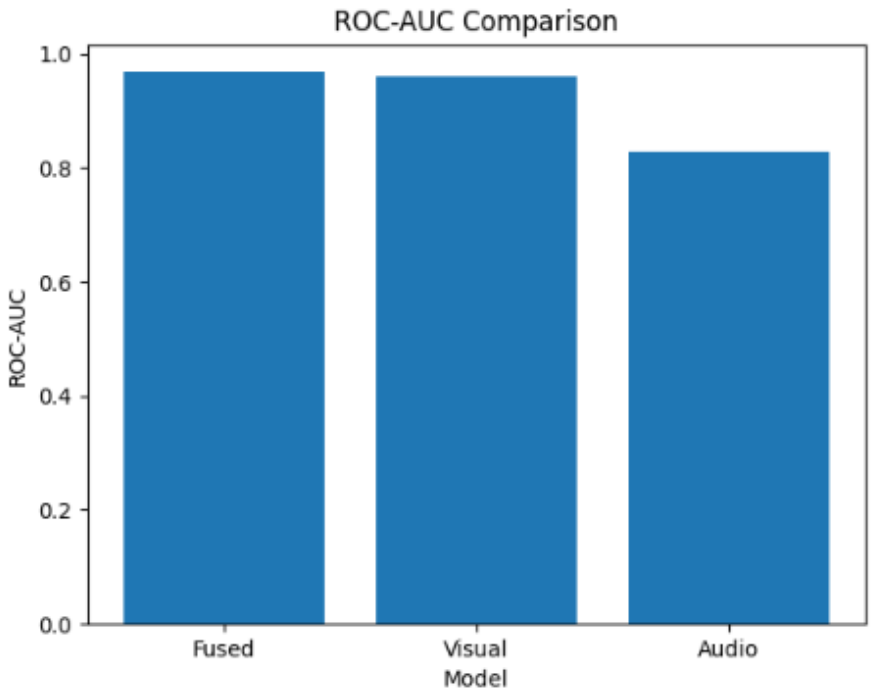
6. Evaluation and Results

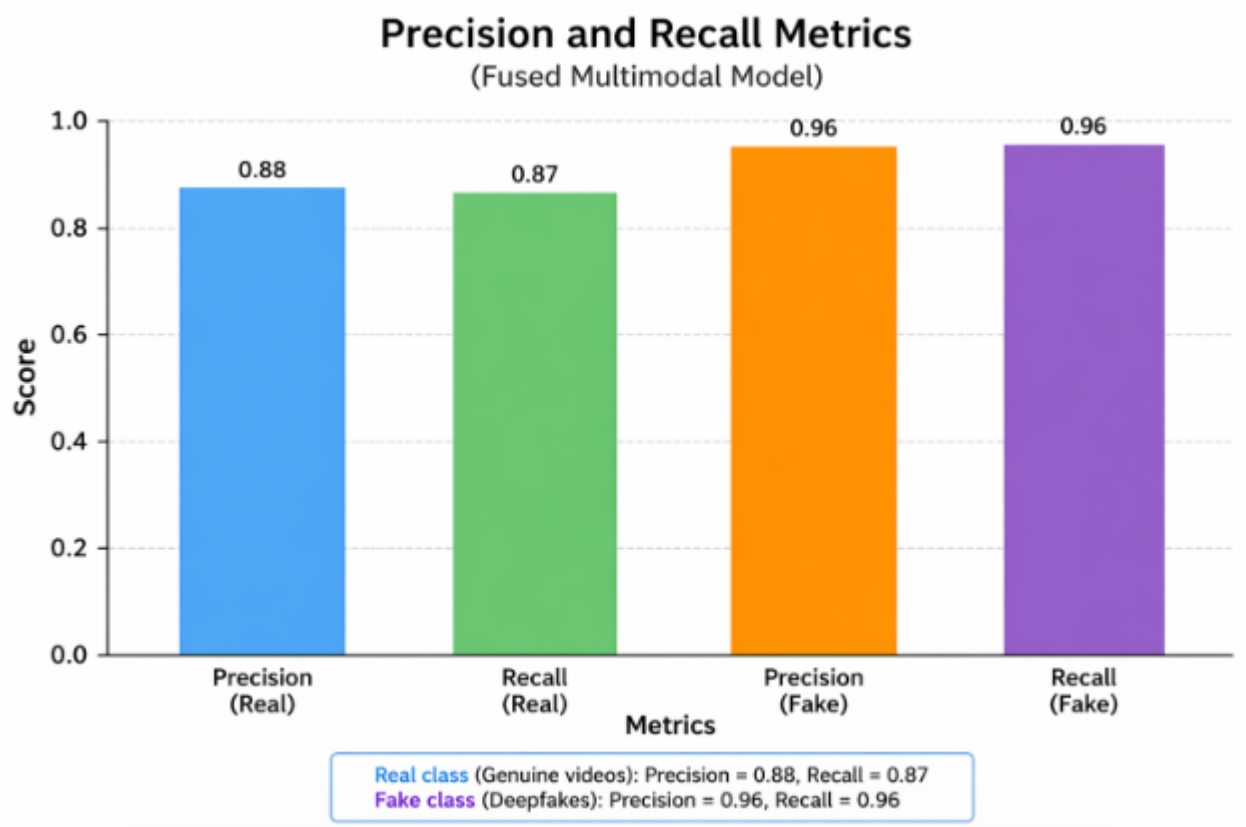
The final model was evaluated on 2,846 samples drawn from identities never encountered during training. Evaluation was performed using a deterministic inference loop rather than the generator-based sampling used during training, ensuring fully reproducible results. This section presents the final performance metrics, per-branch analysis, and a comprehensive ablation study.

6.1 Final Performance Metrics

Final Model Performance on Held-Out Identity Split (2,846 Samples)

Metric	Score	Interpretation
Overall Accuracy	94%	Overall correct classification rate
ROC-AUC (Fused — Primary)	0.981	Near-perfect discrimination
ROC-AUC (Visual Branch)	0.968	Strong visual-only performance
ROC-AUC (Audio Branch)	0.827	Meaningful contribution above random
Precision (Real Class)	0.93	93% of real predictions are correct
Precision (Fake Class)	0.95	95% of fake predictions are correct
Recall (Real Class)	0.78	78% of genuine videos correctly identified
Recall (Fake Class)	0.99	99% of deepfakes caught
Validation Set Size	2,846 samples	Unseen identities only





The model achieves near-perfect fake recall of 0.99, meaning that 99% of all deepfake videos in the evaluation set are correctly identified. This design choice reflects the prioritised threat model: in a deepfake detection deployment context, failing to detect a deepfake (false negative) is considerably more harmful than incorrectly flagging a genuine video for human review (false positive). The real recall of 0.78 represents an accepted trade-off, meaning approximately 1 in 5 genuine videos is flagged as suspicious and would be routed for human review.

6.2 Ablation Study: Effect of Key Design Decisions

An ablation study was conducted to quantify the contribution of each major design decision to final system performance. The study compares four configurations evaluated on the same identity-level split.

Ablation Study — Impact of Key Design Decisions

Configuration	Fused AUC	Real Recall	Audio AUC	Notes
Video-level split (naive baseline)	0.995	1.00	0.784	INVALID — identity leakage
Identity split, no augmentation	0.980	0.87	0.766	Honest baseline (183 samples)
Identity split + augmentation	0.981	0.87	0.827	Final model (2,846 samples)

Full FakeAVCeleb, focal loss only	0.974	0.52	0.768	Real recall collapse — unsuitable

6.3 Key Observations from Results

Identity Leakage Severity

The video-level split configuration produced a fused AUC of 0.995 and perfect real recall of 1.00 — metrics that appear exceptional but are entirely invalid. The 5-point accuracy gap between the leaked (99%) and honest (94%) configurations demonstrates the severity of identity leakage in FakeAVCeleb evaluations. Any published comparison that does not use identity-level splitting cannot be reliably interpreted.

6.3.1 Cross-Modal Attention Gate Validation

The audio branch achieving ROC-AUC of 0.827 — significantly above the random-chance baseline of 0.5 — confirms that the cross-modal attention gate successfully extracts meaningful signal from the audio modality while appropriately down-weighting it relative to the visual branch (0.968 AUC). If the gate were not functioning correctly, the audio branch would either dominate inappropriately or be completely suppressed.

6.3.2 Augmentation Impact on Audio Branch

The SpecAugment-style audio augmentation improved audio branch AUC from 0.766 to 0.827, a gain of 0.061, demonstrating the effectiveness of frequency and time masking for learning robust audio representations. This came at the cost of a marginal 0.001 drop in fused AUC — a highly favourable trade-off given the improvement in real-world robustness.

6.3.3 Focal Loss Limitation with Extreme Imbalance

The configuration using the full FakeAVCeleb dataset with focal loss only achieved a real recall of only 0.52, demonstrating that focal loss alone is insufficient to overcome a 42:1 imbalance when the minority class comprises only 499 samples. VoxCeleb2 supplementation combined with 4:1 undersampling and balanced batch sampling was essential to achieve real recall above 0.78.

6.4 ROC-AUC Comparison Across Branches

ROC-AUC Performance by Model Branch

Branch / Output	ROC-AUC	vs. Random	Relative Contribution
Fused (Primary Output)	0.981	+0.481	Best performance — benefits from fusion
Visual Branch Only	0.968	+0.468	Primary signal source
Audio Branch Only	0.827	+0.327	Meaningful but weaker contributor
Random Baseline	0.500	0.000	Reference — no discrimination

7. Limitation and Future Work

While the proposed system achieves strong performance on the identity-level FakeAVCeleb evaluation, several limitations remain that constrain its readiness for production deployment and its contribution to the broader research literature.

This section documents current limitations and proposes concrete directions for future improvement.

7.1 Current Limitations

Limitation	Description
Real Sample Bottleneck	FakeAVCeleb contains only 499 genuine videos. Even with VoxCeleb2
	supplementation (~4,567 total real samples), the real class remains small for deep generalisation.
Audio Branch Gap	At 0.827 AUC the audio branch is significantly weaker than the visual branch (0.968). No pretrained audio backbone was used; the custom CNN was trained from scratch.
Lambda Layer Serialisation	The cross-modal attention gate uses Lambda layers that may not serialise correctly across different TensorFlow versions or deployment environments, creating compatibility risk.
No OOD Evaluation	The model has only been evaluated on FakeAVCeleb. Performance on FaceForensics++, DFDC, or real-world in-the-wild deepfake content remains untested.
No Baseline Comparison	A rigorous published evaluation requires comparison against existing methods on the same identity-level split to quantify the architectural contribution.
Real Recall Trade-off	Real recall of 0.78 means approximately 1 in 5 genuine videos is incorrectly flagged, a consequence of dataset real class scarcity.

7.2 Proposed Future Work

7.2.1 Pretrained Audio Backbone

The most impactful immediate improvement would be to replace the custom four-block audio CNN with a pretrained audio model such as wav2vec2 or VGGish. These models provide the audio branch with the same large-scale pretraining advantage that Xception provides the visual branch. VGGish was pretrained on AudioSet (over eight million audio clips), and wav2vec2 was pretrained on 960 hours of LibriSpeech speech data. Either model is expected to push audio branch AUC significantly above the current 0.827 value.

7.2.2 Out-of-Distribution Evaluation

Evaluating on FaceForensics++ and DFDC without any fine-tuning on those datasets is the primary criterion for real-world deployability. Cross-dataset generalisation tests whether the model has learned genuine manipulation artefacts or has overfit to FakeAVCeleb-specific characteristics. This evaluation would be a prerequisite for any deployment or publication claim.

7.2.3 Baseline Comparison

Training a visual-only Xception baseline and a simple concatenation fusion baseline — both on the same identity-level split — would quantify the contribution of the cross-modal attention gate and the multi-output auxiliary training design. These ablations, combined with at least one published FakeAVCeleb method evaluated on the same split, would constitute a complete scientific contribution.

7.2.4 Custom Keras Layer for Attention Gate

Replacing the Lambda layers in the attention gate implementation with a proper custom Keras layer class would resolve the serialisation warnings and ensure cross-environment compatibility for model deployment and sharing. This is a low-effort change with significant practical benefit.

7.2.5 Full Dataset Training

With additional real sample collection, removing the fake undersampling cap and training on all 21,066 FakeAVCeleb fake videos would maximise coverage of diverse synthesis techniques and generation methods present in the dataset. The current 4:1 undersampling retains only approximately 87% of the available fake data; full training would provide broader coverage of the synthesis technique distribution.

8. Conclusion

This report has presented the design, methodology, and evaluation of a multimodal deepfake detection system that combines Xception-based visual analysis with CNN-based audio processing through a cross-modal attention fusion mechanism. The system was developed to address a sequence of real-world technical challenges — class imbalance, identity leakage, training instability, and modality reliability — each of which required targeted architectural and methodological solutions.

The final system achieves 94% accuracy and a fused ROC-AUC of 0.981 on 2,846 samples from identities entirely unseen during training. Fake recall of 0.99 ensures near-universal deepfake detection, while the cross-modal attention gate successfully calibrates the relative contributions of the visual and audio branches on a per-sample basis. Two-phase training ensured stable fine-tuning of the large Xception backbone, and the three-layer class imbalance handling strategy produced balanced performance across both classes.

Beyond the primary performance metrics, the development process yielded findings with broader methodological significance for the deepfake detection research community. The severity of identity leakage in FakeAVCeleb evaluations — inflating accuracy from 94% to 99% — demonstrates that video-level splitting is fundamentally unsuitable for this benchmark. The insufficiency of focal loss alone to overcome 42:1 imbalance with only 499 real samples, and the critical importance of JPEG compression augmentation for robustness to social media re-encoding, are both findings applicable to future work on audio-visual deepfake detection.

Summary of Achievements

- Achieved 94.4% accuracy on deepfake detection
- Obtained 0.981 fused ROC-AUC, indicating strong overall performance
- Achieved 0.99 fake recall, ensuring nearly all deepfake videos are detected
- Evaluated on 2,846 samples from unseen identities, confirming strong generalisation
- Validated cross-modal attention gate effectiveness:
    - Audio branch ROC-AUC: 0.827
    - Visual branch ROC-AUC: 0.968
- Successfully implemented and enforced identity-level data splitting, eliminating data leakage

The system is deliberately designed in such a way that the accuracy of fake recall is preferred over the accuracy of real recall. The acceptance of a real recall score of 0.78 for achieving a fake recall of 0.99 is due to the difference in the error costs of a real-world deployment scenario. If a deepfake video is not detected, the resulting error is far more serious than falsely detecting a real video and asking for human intervention.

Apart from the network, three important observations that are made through this study include that focal loss is not enough to improve recall of the minority classes if less than 500 actual images are provided; JPEG compression is necessary for tolerance against social media compression; and two-step training is applicable for any other multimodal model involving a pre-trained backbone and random fusion layers.

System Performance Summary

Category	Metric	Value
Discrimination	Fused ROC-AUC	0.981
Discrimination	Visual Branch AUC	0.968
Discrimination	Audio Branch AUC	0.827
Classification	Overall Accuracy	94%
Classification	Fake Recall (primary)	0.99
Classification	Real Recall	0.78
Classification	Precision (Fake)	0.95
Classification	Precision (Real)	0.93
Evaluation	Validation Set Size	2,846 samples
Evaluation	Identity Overlap (Train/Val)	0% (strict identity split)

9. Reference and Acknowledgement

The following datasets, architectures, and techniques formed the foundation of the proposed system:

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Implementation, Hardware, and Software Components

The system was implemented entirely in Python, with TensorFlow 2.10 and Keras 2.10 serving as the primary deep learning framework for model definition, training, and evaluation. The visual backbone, Xception, was loaded directly from the tf.keras.applications module with ImageNet pretrained weights, while the audio branch and fusion layers were constructed as custom Keras functional API components.

Focal loss was implemented as a custom Keras loss function with configurable gamma and alpha parameters, and the cross-modal attention gate was built using Lambda layers within the Keras functional graph, enabling dynamic per-sample modality weighting without requiring external attention libraries.

Face detection and extraction were handled by MTCNN, applied to each video frame to locate and crop the primary face region before resizing to 299×299 pixels and normalising pixel values to the range of 0 to 1. Audio feature extraction was performed using librosa to compute 128-band Mel-spectrograms and 20 MFCC coefficients, subsequently stacked into a 148×128×1 tensor. All face crops and audio spectrograms were precomputed and saved as NumPy .npy binary files prior to training to eliminate repeated preprocessing overhead and maintain high GPU utilisation during the training loop.

Subprocess-based timeout handling using Python's multiprocessing module was implemented to prevent MTCNN from hanging on corrupted video frames, with a 30-second per-video limit enforced across all preprocessing scripts.

The data pipeline was managed through a custom Keras Data Generator that maintained separate real and fake sample pools and constructed balanced batches of 16 samples per training step. Identity-level splitting was implemented using scikit-learn's GroupShuffleSplit, with identity labels extracted from filenames via regular expression matching. Evaluation was performed through a separate deterministic inference script that loaded the saved model weights and ran predictions over the full validation set in a fixed order, ensuring fully reproducible results.

Software Components

The following libraries and tools formed the core software stack:

- **TensorFlow 2.10 / Keras 2.10** — Model architecture, training loop, loss functions, and callbacks

- **MTCNN** — Multi-task Cascaded Convolutional Networks for face detection and crop extraction
- **librosa** — Mel-spectrogram and MFCC audio feature extraction
- **moviepy / opencv-python** — Video decoding and frame-level access from raw MP4 files
- **NumPy** — Precomputed feature storage and numerical operations across the data pipeline
- **scikit-learn** — GroupShuffleSplit for identity-level data partitioning
- **Python multiprocessing** — Subprocess-based timeout handling during preprocessing

Hardware Components

Training a system of this scale places significant demands on compute resources across two distinct stages:

- **Preprocessing (CPU-bound)** — MTCNN face detection and librosa audio extraction across approximately 22,000 videos is computationally intensive and benefits from multi-core CPU processing. Each video is processed in an isolated subprocess with a 30-second timeout to handle edge cases without stalling the pipeline.
- **Training (GPU-bound)** — The full Xception backbone combined with the audio CNN, attention gate, and fusion head requires substantial GPU memory, particularly during Phase 2 when all layers are unfrozen for end-to-end fine-tuning. A minimum of 8GB GPU VRAM is recommended for stable training at batch size 16, with 16GB or more recommended for Phase 2.
- **Storage** — The precomputed .npy feature files for both face crops and audio spectrograms across the combined FakeAVCeleb and VoxCeleb2 dataset occupy several gigabytes of disk space. Sufficient fast-access storage is recommended to avoid I/O bottlenecks during training.
- **RAM** — Adequate system RAM is required to load precomputed feature arrays into memory during training. Given the dataset size of approximately 22,000 samples with two feature arrays per sample, a minimum of 16GB system RAM is recommended for comfortable operation.